AI-Enhanced Routing Optimization for SDN/NFV Architectures in Large-Scale IoT Networks

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Abstract— The integration of Software Defined Networking (SDN) and Network Function Virtualization (NFV) with the Internet of Things (IoT) offers scalable, programmable, and efficient network infrastructures. However, traditional routing mechanisms often fail to adapt to the dynamic and heterogeneous nature of large-scale IoT networks. This paper proposes a Deep Q-Network (DQN)-based routing optimization method within SDN/NFV architectures. Our approach considers real-time metrics including latency, bandwidth, memory availability, and processing load to make adaptive routing decisions. Simulation results on a 200-node IoT topology demonstrate significant improvements in Packet Delivery Ratio (PDR), end-to-end latency, and load distribution compared to conventional routing algorithms. The paper also discusses limitations such as training complexity, scalability, and energy consumption, while proposing potential enhancements for future research.

Keywords—AI, IOT, routing, security, sdn, nfv

I. Introduction

The rapid expansion of the Internet of Things (IoT) has resulted in highly heterogeneous, large-scale, and dynamic networks. These environments generate massive data flows that conventional routing protocols, such as OSPF, cannot efficiently handle. Software Defined Networking (SDN) and Network Function Virtualization (NFV) have emerged as promising paradigms to enable flexible traffic management and service deployment. However, they face limitations in scalability, real-time adaptation, and efficient resource utilization when applied to IoT environments.

Artificial Intelligence (AI), particularly Reinforcement Learning (RL), provides adaptive and intelligent mechanisms to optimize routing in dynamic networks.

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Among these, Deep Q-Networks (DQN) leverage neural networks to learn optimal policies for high-dimensional state spaces.

Despite their potential, challenges remain regarding scalability, training complexity, and energy efficiency.

In this work, we propose a DQN-based approach that integrates SDN and NFV to achieve dynamic, adaptive, and scalable routing for IoT networks. The main contributions are as follows:

- A novel multi-objective reward function that jointly considers latency, packet delivery ratio (PDR), hop count, and resource utilization.
- Integration of NFV for dynamic service chaining and VNF placement in SDN-enabled IoT environments.
- Large-scale evaluation with 200+ nodes, implemented using Mininet and Ryu, benchmarked against OSPF, Q-Routing, and load balancing.
- Critical discussion of the benefits, limitations, and potential improvements of the proposed approach.

II. Related Works

Several studies have investigated the use of artificial intelligence techniques to address routing challenges in dynamic networks. Traditional routing algorithms such as Dijkstra and OSPF focus on static metrics like hop count and fail to adapt to real-time network changes, leading to congestion and packet loss.

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Q-Routing [1], one of the earliest RL-based routing strategies, showed promise in dynamic networks but lacked scalability in large deployments. Mao et al. [2] introduced deep reinforcement learning for resource management in cloud data centers, demonstrating the applicability of DRL in network optimization. Wang et al. [3] applied DRL specifically to SDN, integrating deep Q-networks for routing decisions with encouraging results.

Other approaches have used fuzzy logic [4] to handle uncertainty in network metrics, but these often struggle with scalability and complexity. More recent work includes multi-agent reinforcement learning for decentralized routing, yet these solutions introduce challenges in coordination and convergence. Additionally, hybrid AI approaches have been proposed that combine neural networks with traditional heuristics, but they often require extensive feature engineering.

More recent surveys such as [5] and [6] provide comprehensive overviews of DRL in networking, while works like [7] propose meta-learning and transfer learning to address cold-start and convergence issues in real-time environments.

Table 1 provides a comprehensive comparison of existing methods, highlighting their respective strengths and limitations, and motivating the need for a more adaptive and intelligent routing mechanism as proposed in this study.

TABLE I. COMPARISON OF ROUTING APPROACHES IN SDN/NFV-BASED IOT NETWORKS

Approach	**Adaptivity**	**Scalability**	**Real-Time Decision**	**Learning-Based**	**PDR**	**Latency**	**Load Balance**
OSPF [8-10]/Dijkstra[11-13]	Low	High	No	No	Medium	High	Poor
Q-Routing [1][14]	Medium	Low	Partial	Yes	Medium	Medium	Medium
Fuzzy Logic [4]	Medium	Medium	Yes	Partial	Good	Medium	Medium
DRL-SDN (Wang et al.) [3]	High	High	Yes	Yes	Good	Low	Good
Our Proposed DQN Method	High	High	Yes	Yes	Excellent	Very Low	Excellent

As shown in Table 1, classical algorithms like OSPF and Dijkstra offer high scalability but lack adaptivity and learning capabilities, resulting in suboptimal performance in real-time scenarios. Q-Routing and fuzzy logic techniques introduce intelligence but fall short in large-scale or highly dynamic contexts. DRL-based methods have demonstrated superior adaptability and performance, though some suffer from high training overhead and convergence delays[15].

Building upon these findings, our proposed approach leverages deep Q-networks to deliver a scalable and intelligent routing solution tailored for SDN/NFV-enabled IoT environments[16]. It outperforms previous methods in packet delivery ratio (PDR), latency, and load balancing, making it a viable candidate for real-time, large-scale network optimization.

III. PROPOSED ARCHITECTURE

The proposed architecture integrates a Deep Q-Network (DQN) agent within the SDN controller, which interfaces with OpenFlow switches to manage routing in real-time. IoT nodes generate traffic that is monitored continuously by the controller, while NFV components handle virtualized network functions and, where applicable, basic service function chaining. The architecture ensures centralized control while maintaining programmability and scalability.

A. AI Agent Integration

The AI module is embedded directly into the SDN controller. It acts as a decision-making agent that evaluates the state of the network and determines the best routing action for each flow. The DQN agent receives a state vector from each node composed of the following:

- Available bandwidth: Indicates the remaining data capacity on the outgoing link.
- Queue size (congestion): Measures the current buffer occupancy.
- Processing load: Reflects the CPU usage and task handling status of the node.
- Available memory: Indicates free memory for processing and buffering.
- Distance (hop count): Number of hops required to reach the destination node.

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This information forms the basis of the s_t state that the DQN processes.

B. Action and Reward

- **Action**: The DQN outputs the optimal next hop for routing a particular data flow. Each action corresponds to selecting a neighboring node for forwarding.
- **Reward Function**: Designed to guide the agent toward minimizing:
  - End-to-end latency
  - Congestion levels
  - Packet loss
  - And to maximize:
  - Packet delivery ratio
  - Resource efficiency (balanced use of memory and CPU)

Reward Function: The reward function guides the agent to optimize multiple objectives simultaneously:

$$\text{Reward} = \alpha*(-\text{Latency}) + \beta*(\text{PDR}) + \gamma*(-\text{Queue Size}) + \delta*(\text{Load Balance Score})$$

- α (latency weight): emphasizes minimizing end-to-end delay.
- β (PDR weight): prioritizes successful packet delivery.
- γ (queue weight): discourages congestion at nodes.
- δ (load balance weight): encourages even distribution of traffic across the network.

Coefficients (α, β, γ, δ) are tunable parameters chosen based on network priorities and empirical observations to balance latency, reliability, congestion, and fairness in routing decisions.

C. System Flow

The operational workflow proceeds as follows:

1) Monitoring: Real-time traffic and resource metrics are collected from all OpenFlow switches.
2) State Formation: Metrics are aggregated into state vectors for each active flow.
3) DQN Decision: The agent uses the current policy (neural network) to select the optimal next hop.
4) Routing Update: The controller updates forwarding tables in relevant switches using OpenFlow rules.
5) Learning: Experience tuples (state, action, reward, next state) are stored in a replay buffer and used to improve the policy network via stochastic gradient descent.

IV. Simulation and Evaluation

To evaluate the performance of our AI-enhanced routing mechanism, we conducted extensive simulations using Mininet and the Ryu SDN controller. The DQN model was implemented using TensorFlow, with a network topology of 200 IoT nodes structured in a clustered mesh configuration.

A. Simulation Setup

To evaluate the performance of the proposed approach, a comprehensive simulation environment was developed using software-defined networking (SDN) principles. The Ryu SDN controller, implemented in Python, was utilized to manage the control plane logic, while Mininet served as the network emulation tool to replicate realistic network conditions. The experimental setup consisted of 200 IoT nodes organized into 10 clusters to mimic a scalable and distributed IoT infrastructure. Several routing strategies were benchmarked, including OSPF, Load Balancing, and Q-Routing, with a focus on assessing key performance metrics such as Packet Delivery Ratio (PDR), Average End-to-End Latency, Load Balance Index (LBI), Network Throughput, and the CPU/Memory usage on the control plane.

- Controller: Ryu SDN controller (Python-based), running on a server with Intel Xeon 8-core CPU, 32GB RAM, and 1 Gbps NIC.
- Emulation Tool: Mininet, simulating 200 IoT nodes organized into 10 clusters to reflect realistic deployment and stress-test network performance.
- Routing Baselines: OSPF, Load Balancing, Q-Routing.
- Metrics Measured: Packet Delivery Ratio (PDR), Average End-to-End Latency, Load Balance Index (LBI), Network Throughput, CPU/Memory Usage.

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Justification for Simulation Choices:

- The 200-node topology represents a large-scale IoT network, enabling testing of scalability and stress under high traffic conditions.
- DQN training parameters are chosen based on convergence stability and standard reinforcement learning practices:
  - Learning rate = 0.001: ensures smooth gradient updates.
  - Discount factor γ = 0.95: balances immediate and future rewards.
  - Replay memory = 10,000 transitions, batch size = 64: supports effective experience replay.
  - Epsilon-greedy exploration with exponential decay (1.0 to 0.01): promotes exploration early and exploitation later for stable policy learning.

B. *Identify the Headings*

For the reinforcement learning component, a Deep Q-Network (DQN) was employed to optimize routing decisions within the SDN framework. The DQN was trained over 10,000 episodes to ensure stable policy convergence. The neural network architecture consisted of three fully connected layers with ReLU activation functions, designed to effectively capture complex state-action relationships. Key training parameters included a learning rate of 0.001, a discount factor (gamma) of 0.95, and a replay memory capacity of 10,000 transitions to support experience replay. A batch size of 64 was used during training, with exploration managed through an epsilon-greedy strategy featuring exponential decay—starting from 1.0 and gradually decreasing to 0.01.

The specific configuration details of the DQN training process are summarized below:

- DQN trained over 10,000 episodes
- Neural Network: 3 fully connected layers with ReLU activations
- Learning rate: 0.001
- Gamma (discount factor): 0.95
- Replay memory size: 10,000 transitions
- Batch size: 64
- Epsilon-greedy exploration with exponential decay (starting at 1.0, ending at 0.01)

C. *Results and Analysis*

The proposed Deep Q-Network (DQN)-based routing algorithm demonstrated substantial improvements over traditional and baseline AI-based methods in all key performance metrics. We evaluated four approaches: OSPF (traditional), Load Balancing (weighted round-robin), Q-Routing (early RL method), and our Proposed DQN method.

TABLE II. PERFORMANCE COMPARISON ACROSS ROUTING METHODS

Approach	Packet Delivery Ratio (%)	Latency (ms)	Load Balance Index
OSPF	84.1	103	0.65
Load Balancing	88.7	85	0.78
Q-Routing	89.4	78	0.80
Proposed DQN	**96.3**	**58**	**0.87**

Key Observations:

- Packet Delivery Ratio (PDR): The DQN approach achieves the highest PDR (96.3%), demonstrating superior reliability in dynamic traffic conditions.
- Latency: The proposed method reduces end-to-end latency to 58 ms, which is ~44% lower than OSPF and ~25% lower than Load Balancing.
- Load Balance Index (LBI): A higher LBI (0.87) reflects better traffic distribution across the network, reducing bottlenecks and improving QoS.

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FIGURE I: COMPARATIVE CHART FOR OUR PROPOSED SOLUTION

Figure 1 (shown above) visualizes the improvement of our method across the three critical metrics: PDR, latency, and load balancing.

As illustarted in fig1, we have a graphical comparison in:

- PDR Improvement: Our DQN method surpasses others with nearly 12% gain over OSPF.
- Latency Reduction: Nearly 45% lower latency than OSPF and ~25% lower than Load Balancing.
- Load Balance: DQN achieves more uniform traffic distribution.

And as an Additional Observations:

- Control Overhead: Slightly increased due to inference computations, but remains within acceptable operational bounds.
- Throughput: Peak throughput achieved was 9.6 Gbps with DQN, outperforming OSPF's 7.1 Gbps.
- Convergence: The DQN agent converges after 3,000 episodes, indicating fast adaptability.

These results confirm the effectiveness of AI-enhanced routing strategies. The DQN agent continuously learns from network feedback and adapts routing decisions in real-time, outperforming static or heuristic-based solutions, especially in large-scale and heterogeneous IoT environments.

V. DISCUSSION

The results underscore the robustness and adaptability of the proposed DQN-based routing framework. By continually learning from real-time network states, the agent successfully avoids congested paths and balances the load across the network. Compared to static or rule-based routing, the DQN agent adapts dynamically, showing high responsiveness to topological changes and fluctuating traffic.

However, the system still requires a balance between training complexity and inference speed. In larger topologies, the dimensionality of the state space grows, which can impact convergence speed. Model pruning, transfer learning, and hierarchical RL could be explored to mitigate these effects. Moreover, the centralized nature of the SDN controller may introduce a bottleneck; distributing inference tasks to edge nodes is a potential enhancement.

Compared to Wang et al. [17], our method integrates additional state features, including memory and hop count, along with a multi-objective reward function, which enhances adaptability in large-scale IoT networks. Other recent DRL-based routing approaches, such as dueling DQN and PPO-based multi-path routing, also aim to improve adaptability, but our approach demonstrates competitive or superior performance in terms of Packet Delivery Ratio (PDR) and latency while maintaining a simpler single-agent implementation suitable for centralized SDN controllers. However, there are limitations: DQN training over 10,000 episodes is computationally intensive, which may pose constraints in very large networks. Centralized control could become a bottleneck in networks with thousands of nodes, suggesting that edge or hierarchical RL strategies may be necessary for scaling. Additionally, the computational overhead of DQN inference contributes to increased energy consumption,

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which is a concern for resource-constrained IoT devices. Despite these challenges, control overhead remains acceptable, peak throughput with DQN reached 9.6 Gbps, outperforming OSPF's 7.1 Gbps, and the DQN policy converges after approximately 3,000 episodes, indicating efficient learning.

VI. Conclusion and Future Work

In this paper, we presented a novel AI-based routing optimization technique for SDN/NFV-enabled IoT networks, leveraging Deep Q-Networks (DQN) to make adaptive routing decisions based on real-time network metrics such as latency, bandwidth, memory, and processing load. Our method demonstrated significant improvements in key performance indicators, achieving a Packet Delivery Ratio of 96.3%, reducing end-to-end latency to 58 ms, and enhancing load balance across the network compared to traditional routing methods. While the approach shows strong adaptability and performance, limitations include high training complexity, centralized control constraints, and increased computational resource requirements. Future work will focus on energy-aware routing to optimize efficiency for IoT devices, exploring multi-agent reinforcement learning for distributed control, and integrating the solution with emerging 6G network architectures to enhance scalability and applicability in next-generation IoT environments.

Future research will investigate:

- Federated reinforcement learning for distributed scalability.
- Online learning to allow continual adaptation without retraining from scratch.
- Energy-aware routing extensions to improve sustainability in IoT applications.
- Deployment in real-world testbeds to validate simulation outcomes.

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